



## MySQL – Interview Questions

1. What is the difference between DELETE, TRUNCATE, and DROP?

Cross Q1: Which command can be rolled back?

Cross Q2: What happens to the table structure when you use TRUNCATE?

2. What are primary keys and foreign keys?

Cross Q1: Can a table have multiple primary keys?

Cross Q2: How does a foreign key maintain referential integrity?

3. What are the different types of JOINS in SQL?

Cross Q1: What is the difference between INNER JOIN and LEFT JOIN?

Cross Q2: Can you give a real-time use-case where RIGHT JOIN is used?

4. What is normalization?

Cross Q1: Explain 1NF, 2NF, and 3NF with examples.

Cross Q2: What is denormalization and when is it useful?

5. What is the difference between WHERE and HAVING clause?

Cross Q1: Can you use HAVING without GROUP BY?

Cross Q2: How are filters applied in query execution order?

6. What are aggregate functions in MySQL?

Cross Q1: Name 5 aggregate functions.

Cross Q2: Can you apply aggregate functions without using GROUP BY?

7. What is the difference between CHAR and VARCHAR?

Cross Q1: Which one is better for fixed-length fields like PIN codes?

Cross Q2: How does MySQL handle storage allocation for both?

8. How do indexes work in MySQL?

Cross Q1: What is the impact of indexing on performance?

Cross Q2: What is the difference between clustered and non-clustered index?

9. What is the use of AUTO\_INCREMENT in MySQL?

Cross Q1: Can we reset the value of an AUTO\_INCREMENT column?

Cross Q2: What will happen if we insert a custom value manually?

10. What is a view in MySQL?

Cross Q1: Can we update data using a view?

Cross Q2: How do views help in security and abstraction?

11. What is the difference between IN and EXISTS?

Cross Q1: Which one is better for performance?

Cross Q2: What happens when the subquery returns an empty result?



12. What are constraints in MySQL?

Cross Q1: Name different types of constraints.

Cross Q2: How does UNIQUE differ from PRIMARY KEY?

13. What is a transaction in MySQL?

Cross Q1: What are the properties of a transaction (ACID)?

Cross Q2: How is COMMIT different from ROLLBACK?

14. What is a stored procedure?

Cross Q1: How is a procedure different from a function?

Cross Q2: Can you call a stored procedure from Java using JDBC?

15. What is the difference between IS NULL and = NULL?

Cross Q1: What will WHERE column = NULL return?

Cross Q2: How do you select records with missing data?

16. How do you write a subquery in MySQL?

Cross Q1: Can subqueries be nested?

Cross Q2: What is a correlated subquery?

17. What is the purpose of GROUP BY clause?

Cross Q1: Can you group by multiple columns?

Cross Q2: What happens if you use a column in SELECT that is not in GROUP BY or aggregate?

18. What is the default storage engine in MySQL?

Cross Q1: Compare InnoDB and MyISAM.

Cross Q2: Which engine supports transactions and foreign keys?

19. How do you prevent SQL injection?

Cross Q1: What are prepared statements?

Cross Q2: How does using ORM (like Hibernate) protect against injection?

20. What is the difference between UNION and UNION ALL?

Cross Q1: Which one eliminates duplicate records?

Cross Q2: What is the performance difference between the two?

21. How do you handle duplicate values in a table?

Cross Q1: How can you delete duplicate rows but keep one?

Cross Q2: How to avoid inserting duplicates using constraints?

22. How do you retrieve the top 5 highest salaries from an employee table?

Cross Q1: What will happen if multiple employees have the same 5th highest salary?

Cross Q2: How to retrieve nth highest salary using subquery?

23. How do you export and import MySQL databases?

Cross Q1: What is the use of mysqldump?



Cross Q2: How do you take a backup of only table structure without data?

24. What is the difference between INNER JOIN and SELF JOIN?

Cross Q1: How is SELF JOIN helpful for hierarchical data?

Cross Q2: Can you give a real example using employee-manager table?

25. How do you optimize a slow-running query?

Cross Q1: What is EXPLAIN used for?

Cross Q2: How can indexes help? When can they make performance worse?

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